Muscle development and matrix composition change on different schedules

a Independent roles of the assays

snRNA-seq · 2 libraries per age

All subunits expressed in both libraries;
CM receptors and source identity

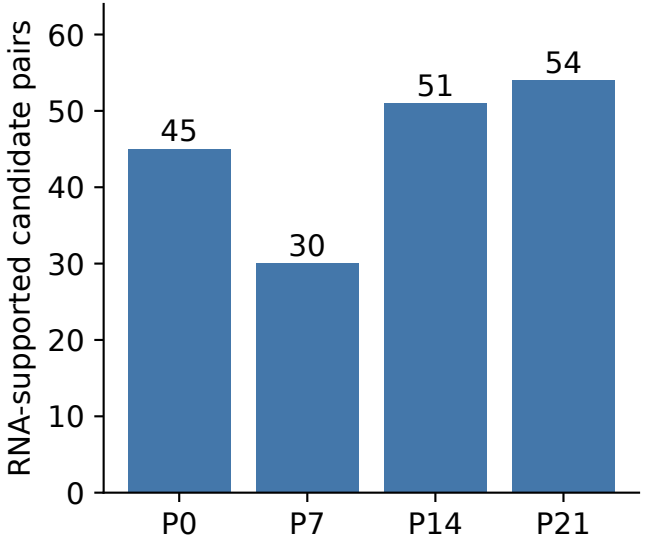
Xenium · observed RNA and locations

Local source expression, CM receptors,
21-gene and component outcomes

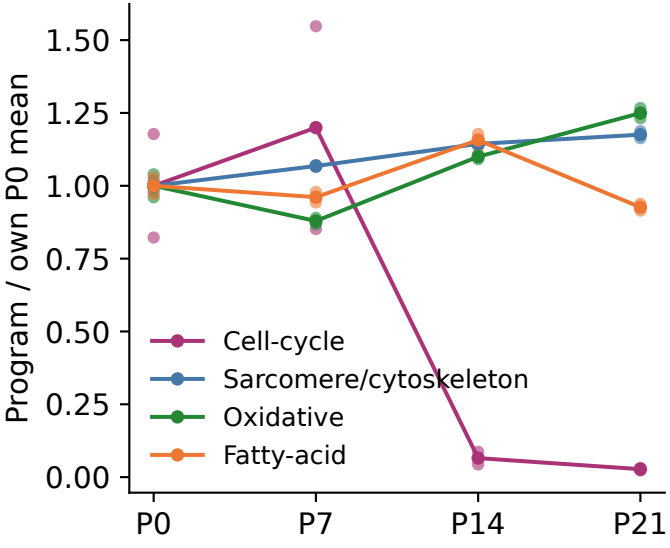
Four buffered test domains

No shared source cells across folds;
≥400 µm between training and test CMs

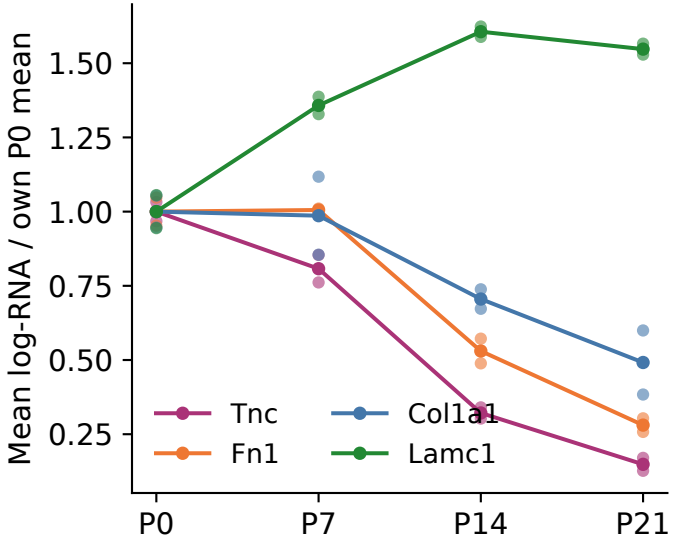
b RNA-supported candidate pairs



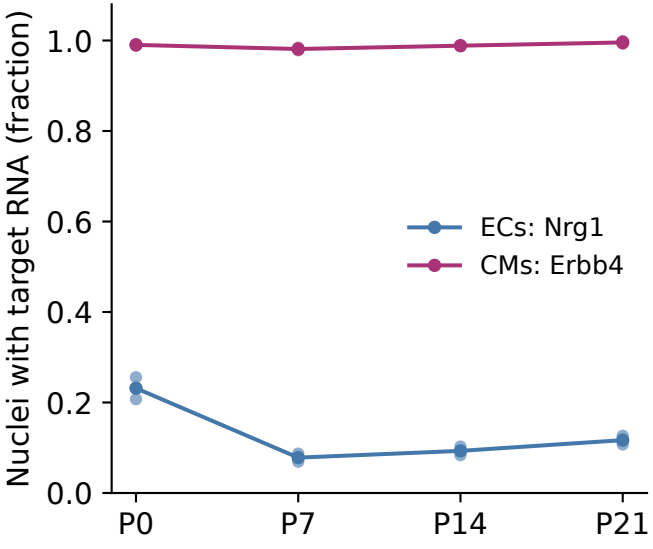
c CM programs change differently



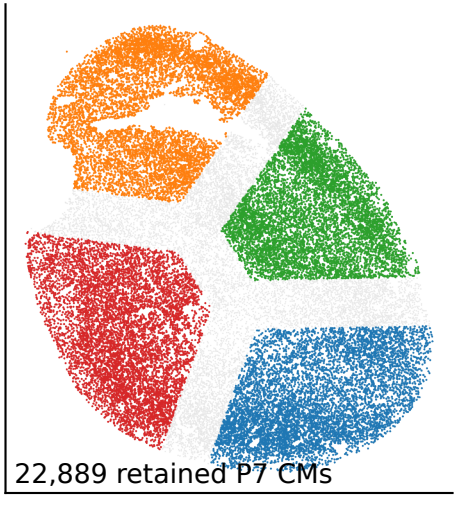
d Fibroblast matrix expression changes

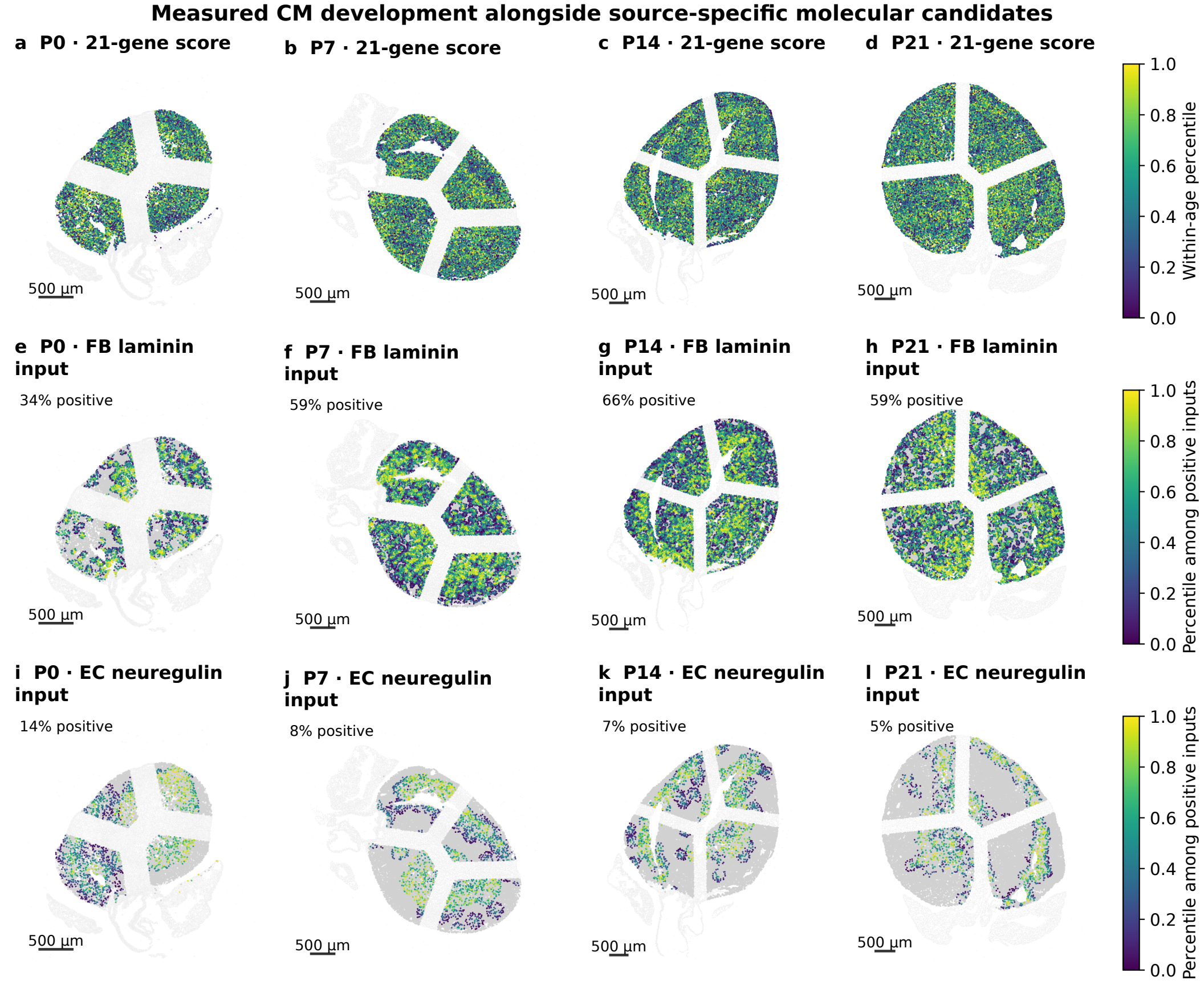


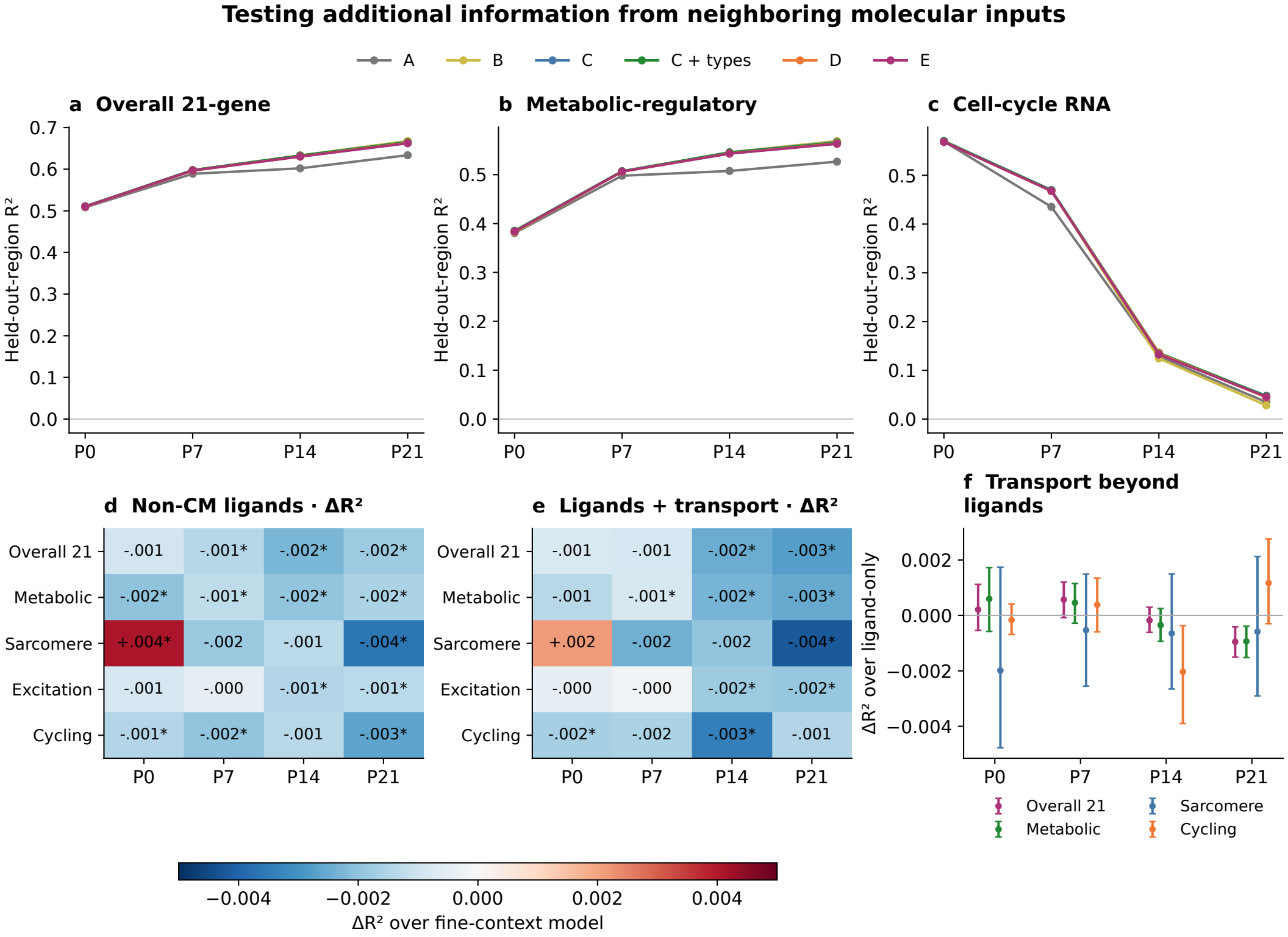
e A vascular candidate has CM support

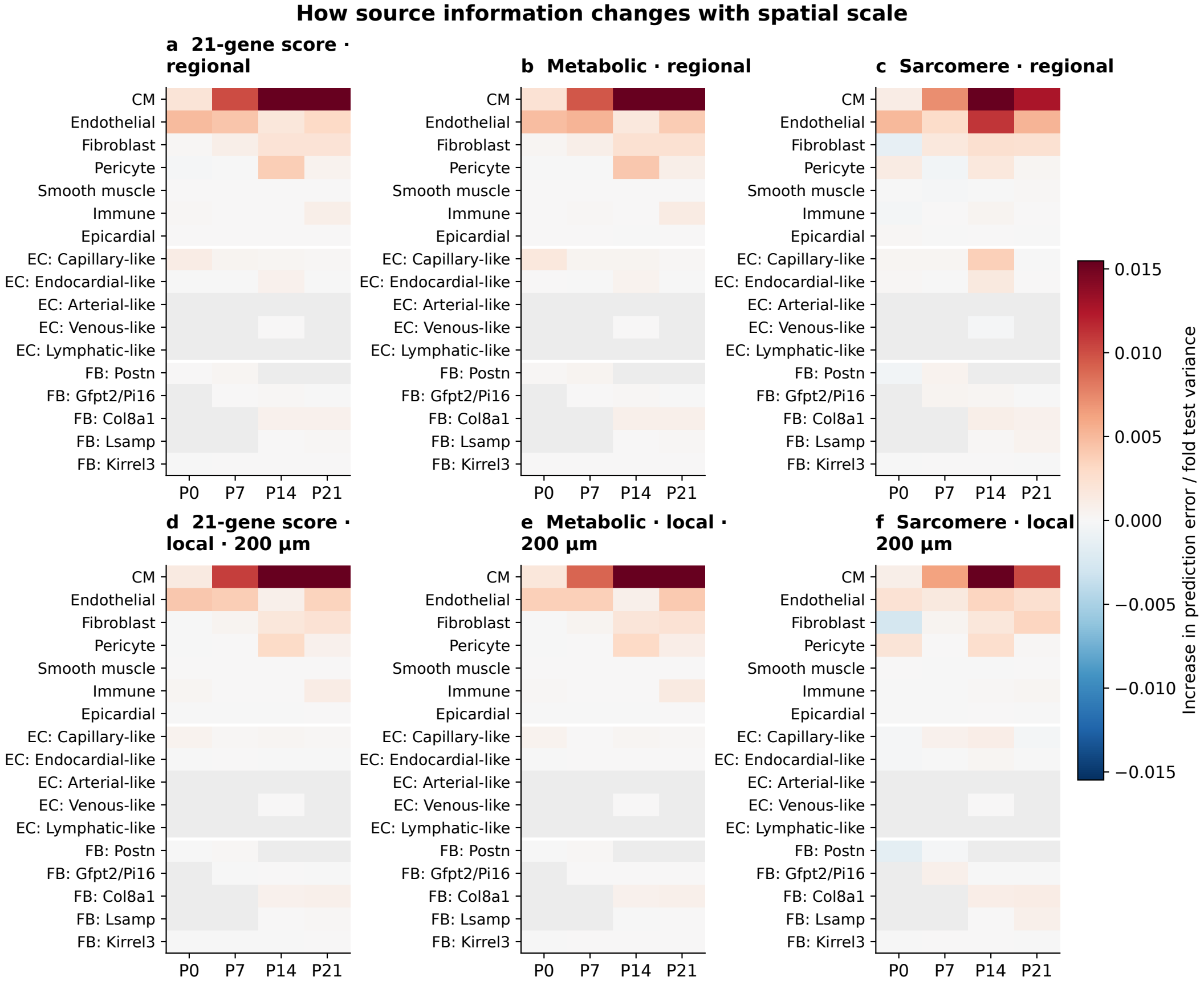


f P7 · buffered domains




Measured CM development alongside source-specific molecular candidates
a P0 · 21-gene score
b P7 · 21-gene score
c P14 · 21-gene score
d P21 · 21-gene score
500 µm
Within-age percentile
1.0
0.8
0.6
0.4
0.2
0.0
e P0 · FB laminin input
34% positive
f P7 · FB laminin input
59% positive
g P14 · FB laminin input
66% positive
h P21 · FB laminin input
59% positive
Percentile among positive inputs
i P0 · EC neuregulin input
14% positive
j P7 · EC neuregulin input
8% positive
k P14 · EC neuregulin input
7% positive
l P21 · EC neuregulin input
5% positive


Testing additional information from neighboring molecular inputs
A
B
C
C + types
D
E
a Overall 21-gene
b Metabolic-regulatory
c Cell-cycle RNA
Held-out-region R²
P0
P7
P14
P21
d Non-CM ligands · ΔR²
e Ligands + transport · ΔR²
Overall 21
Metabolic
Sarcomere
Excitation
Cycling
-.001
-.001*
-.002*
-.002*
-.002*
-.001*
-.002*
-.002*
+.004*
-.002
-.001
-.004*
-.001
-.000
-.001*
-.001*
-.001*
-.002*
-.001
-.003*
-.001
-.001
-.002*
-.003*
-.001*
-.001*
-.002*
-.003*
+.002
-.002
-.002
-.004*
-.000
-.000
-.002*
-.002*
-.002*
-.002
-.003*
-.001
f Transport beyond ligands
ΔR² over ligand-only
Overall 21
Metabolic
Sarcomere
Cycling
−0.004
−0.002
0.000
0.002
0.004
ΔR² over fine-context model


How source information changes with spatial scale
a 21-gene score · regional
b Metabolic · regional
c Sarcomere · regional
d 21-gene score · local · 200 µm
e Metabolic · local · 200 µm
f Sarcomere · local 200 µm
CM
Endothelial
Fibroblast
Pericyte
Smooth muscle
Immune
Epicardial
EC: Capillary-like
EC: Endocardial-like
EC: Arterial-like
EC: Venous-like
EC: Lymphatic-like
FB: Postn
FB: Gfpt2/Pi16
FB: Col8a1
FB: Lsamp
FB: Kirrel3
P0
P7
P14
P21
0.015
0.010
0.005
0.000
−0.005
−0.010
−0.015
Increase in prediction error / fold test variance